AI LAB SERIES

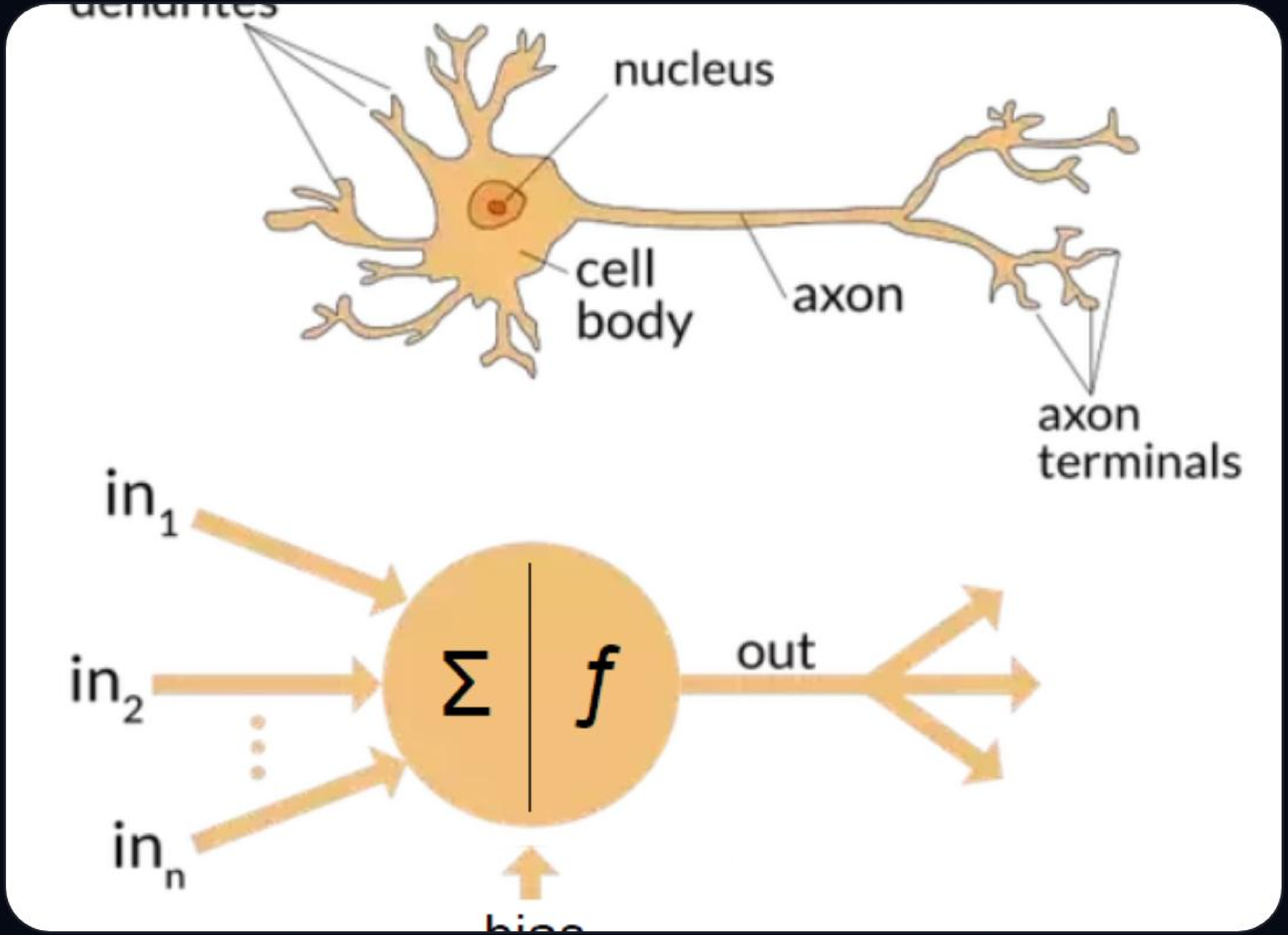
# Perceptron Learning

A Comprehensive Study on the Fundamental Building Block of Neural Networks. Understanding Linear Classifiers through the Lens of Bio-Mimicry.

Air University

Lab 08

# Nature Meets Computation




### The Biological Blueprint

Just as a **Biological Neuron** receives electrical pulses via Dendrites and fires through an Axon, the **Perceptron** takes digital inputs and fires a binary signal.

- **Dendrites** = Input Features ($x_i$)
- **Synapse Strength** = Weights ($w_i$)
- **Axon Firing** = Output ($Y$)

# The Math Behind Decisions

The perceptron acts as a linear combination unit. It sums up all weighted inputs and adds a bias to shift the decision boundary.

$$Y = fb + \sum_{i=1}^{n} w_i x_i$$

*Where $f$ is the Hardlimiter activation function.*

### Key Components

**W:** Weights signify the importance of each feature.

**b:** The Bias allows the model to adjust independent of inputs.

# How Machines "Learn"

### The Update Rule

Learning happens when the model makes a mistake. We shift the weights to be "less wrong".

$$\Delta w_i = ( y_{actual} - y_{predicted} ) \cdot x_i$$

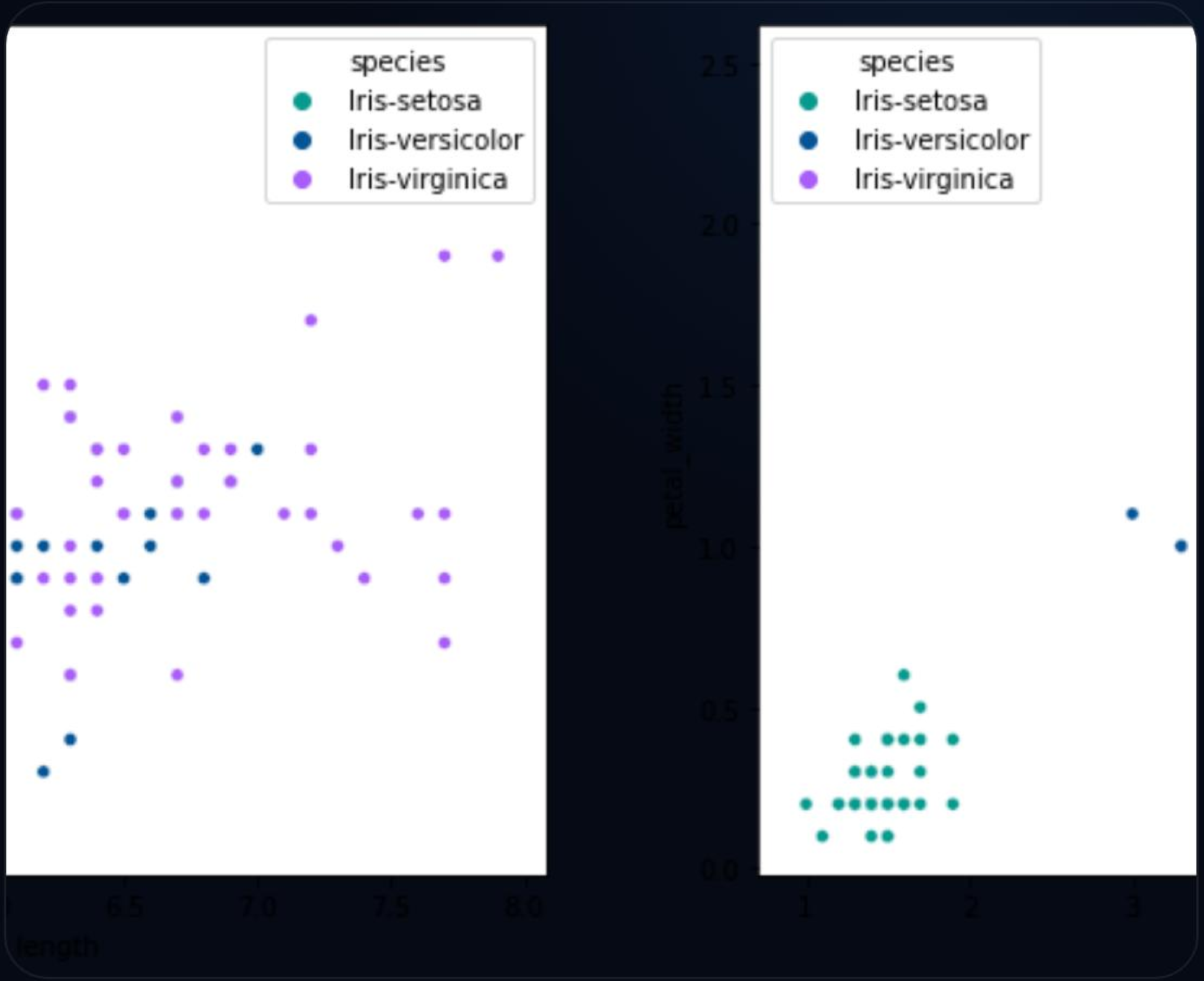
**Algorithm Steps:**

1. Initialize weights and bias to **Zero**.
2. For each sample, calculate predicted output.
3. If prediction is wrong, calculate **Delta**.
4. Update weights and repeat until convergence.

# Linear Separability (Iris)

In the Iris Dataset, **Setosa** and **Versicolor** are linearly separable. This means a single straight line can divide them perfectly.

Our goal is for the Perceptron to find the mathematical equation for this dividing line.



# The Interactive Lab

No manual needed. Test the logic live below.

# Perceptron Simulator

### Input Features

Feature $x_1$ (e.g. Petal Length)

1

Feature $x_2$ (e.g. Sepal Length)

1

Weight $w_1$

0.5

Bias $b$

-0.8

RUN CALCULATION

Waiting for Input...

The hardlimiter threshold is 0.

# Lab Task: Boolean OR

Your task is to design a perceptron that mimics an **OR Gate**. This requires finding weights that satisfy the truth table.

| $x_1$ | $x_2$ | Target $Y$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

**Objective:** Prove misclassification is zero after training. Submit your Python code showing the iterations.

Hint: Start with $w_1=0.5$, $w_2=0.5$, $b=-0.2$ and observe the behavior.

# Accuracy vs Iterations

As the number of iterations increases, the perceptron updates its weights, and the number of misclassified samples decreases towards zero.

AI Circuitry Abstract

# Impact of Features

What happens if we reduce features from 4 to 2?

While 4 features give more "evidence", if the data is highly separable (like Iris), 2 features (Petal Length & Width) are often sufficient for 100% accuracy.

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However, reducing features in complex datasets can lead to "Over-simplification" where the line can no longer separate the classes.

# Lab Deliverables

**Accuracy Plots**

Show how error drops across 10 iterations.

**OR Logic Code**

Verify that your final weights satisfy all OR samples.

# Questions?

Feel free to experiment with the simulator in Tab 7.

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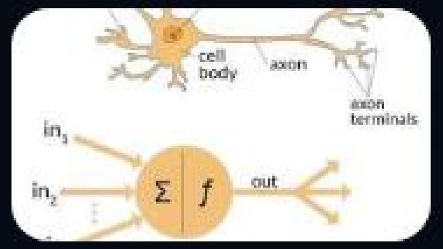
# Image Sources

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Source: medium.com

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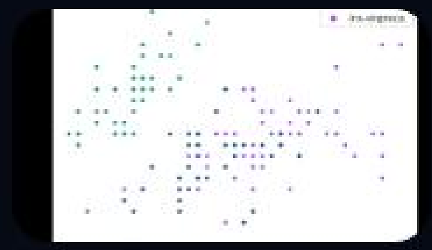

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